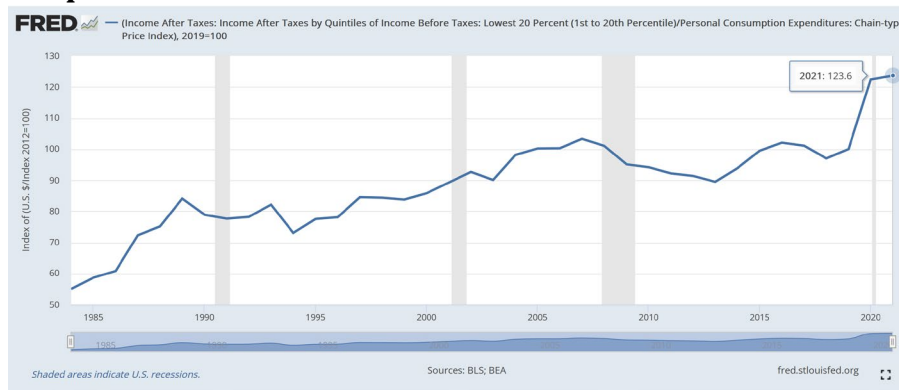


Chapter 4 Examples



After-tax real wages for the bottom quintile of the US income distribution have soared relative to pre-Pandemic levels.

1. Graph the partial equilibrium household utility maximization problem from Chapter 4 assuming, as in the case shown in the above Figure, a large increase in the after-tax real wage.

In your graph, show the initial equilibrium point, the final equilibrium point, as well as the out-of-equilibrium point that represents the pure substitution effect of the change in $w(1-t)$ and the out-of-equilibrium point that represents the pure income effect of the change in $w(1-t)$.

2. Do you think that the bottom quintile of the US income distribution has felt an overall incentive to work more or to work less as a result of these after-tax real wage changes? Does your intuition rely on assumptions about features of this particular income group? Why or why not?

3. Solve for the equilibrium value of labor in the following model economy:

$$U(C, l) = a \ln C + b \ln(l)$$

$$C = w(1 - t)(16 - l)$$

4. Now solve for both C and N in the same economy as in Question 3, where
 $a = .5$; $b = .25$; $w(1-t) = 10$

5. Reflect on your answer to questions 3 and 4: What have you learned about the way the weights in the log-linear utility function impact equilibrium outcomes? Do you predict this will be generally true of all other types of utility functions?

$$U(C, l) = a \ln C + b \ln(l)$$

$$C = w(1 - t)(16 - l)$$

6. Women's labor force participation rates increased dramatically between 1945 and 1995. Using the model from Question 3 (above), could this behavioral change be explained by either or both of the following factors and, if so, how would it be represented in the mathematical model? how would it be graphed?
- Rising after-tax real wages
 - Changes in preferences